

Andy Deng

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SUMMARY

Backend software engineer with 2 years of experience building data-intensive applications, distributed systems, and scalable API platforms. Expertise in asynchronous processing, idempotent system design, and high-concurrency architectures to power reliable data flows, event-driven services, and LLM orchestration at scale.

SKILLS

Languages: Python, TypeScript, Java, Ruby, SQL

Architecture & Data: Distributed Systems, Microservices, Asynchronous Processing, Idempotency, Message Queues

Databases & State: PostgreSQL, Redis, Celery, Data Caching, Server-Side Clustering

Infrastructure: AWS (EC2, S3, RDS, Lambda), Docker, CI/CD (GitHub Actions), RESTful APIs, GraphQL, WebSockets, Karate (Test Suites)

EXPERIENCE

Pegasystems | Software Engineer (Backend Infrastructure) Boston, MA | May 2025 – Present

- Build backend services end-to-end for Pega Blueprint, an agentic workflow platform supporting **2M+** enterprise workflows
- Built LLM abstraction layer with provider DI, structured output parsing, and centralized retry/fallback, enabling multi-model routing across Blueprint generation pipelines
- Parallelized LLM calls across Blueprint generation stages (data model, cases, personas), cutting end-to-end generation time over **50%**
- Migrated Blueprint generation status from client polling to WebSocket event streams, eliminating rate-limit and timeout failures and cutting backend load **30%+**
- Cut Blueprint initial load from **6s to 1.5s** by parallelizing a sequential API waterfall and fixing N+1 query patterns across orchestration services
- Designed CI/CD pipelines (GitHub Actions) for zero-downtime deployments across critical orchestration services
- Built behavior-driven end-to-end integration test coverage for core APIs using Karate

forREAL | Software Engineer (Frontend / Backend / Data) Boston, MA | Sep 2024 – May 2025

- Built idempotent Stripe payment pipeline (Django/Celery on AWS EC2/RDS) processing **\$120k+** in monthly rent across autopay, lease signing, and identity verification, with webhook signature verification and reconciled state machines
- Designed Django data models and APIs for lease lifecycle, role-based access, and document signing flows, backed by an integration test suite covering payment and webhook edge cases
- Cut external API latency and Google Maps costs **70%** via server-side clustering, distributed caching, and lazy loading for geospatial queries

Vanta | Software Engineer Intern (Backend API) Remote | Aug 2023 – Feb 2024

- Built Redis pub/sub notification system routing 10k+ daily chat payloads, saving \$2k/month and **80%** of projected Stream Chat costs
- Built idempotent Stripe payment flows for 1k+ orders, eliminating duplicate-charge edge cases under retry conditions

PROJECTS

Remes (remes.so): Distributed API Pipeline

- Built end-to-end pipeline coordinating LLM extraction, streaming SSE, and third-party enrichment APIs with parallel execution, abort control, and graceful degradation
- Designed strongly-typed REST APIs with Zod validation, auth middleware with rate limiting, and async workflow coordination

Sigdiff: AST Analysis CLI

- Built zero-dependency npm CLI to detect breaking TypeScript API changes via AST analysis, shipping at ~8KB
- Designed a four-stage pipeline (extract, diff, classify, format) on top of the TypeScript Compiler API

EDUCATION

Boston College | Bachelor of Arts in Computer Science (GPA: 3.7)

Chestnut Hill, MA | May 2024